When Speed Slows You Down:

How Cumulative Workload Impacts Fatigue and Self-Regulation in Gig Delivery

Final Report – Seminar in Behavioral Supply Chain Management

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1. Introduction

1.1 General Background & Conflict

The rapid emergence of quick-commerce platforms – *defined here as app-mediated delivery services guaranteeing orders in under 30 minutes, managed via real-time algorithmic assignment* – has significantly reshaped last-mile delivery logistics, especially in densely populated markets such as India, where the companies promising delivery within mere minutes have captured a substantial share of the $11 billion online grocery market, projected to dominate approximately 70% of a $60 billion market by 2030. This explosive growth reflects consumers' increasing expectations for instantaneous fulfillment but has simultaneously sparked a critical debate on its sustainability. Central to this debate is a significant conflict: the aggressive pursuit of ultra-fast delivery may inadvertently compromise gig delivery workers' safety, performance, and overall well-being due to cumulative workload and associated fatigue.

1.2 Research Question & Motivation

This research specifically addresses this pressing conflict by examining how cumulative workload influences the behavioral performance of gig delivery workers within quick-commerce operations. The core research question guiding this study is: *"How does cumulative workload impact the behavioral performance—specifically, delivery speed and self-regulation behaviors—of gig workers?"* We further investigate two specific behavioral responses: (1) performance deterioration, measured by delivery speed and deadline misses, and (2) self-regulation, assessed by voluntary breaks and early log-off rates.

1.3 Previous Studies & Research Gap

Prior studies in behavioral operations management and occupational health provide evidence that prolonged workload and resultant fatigue adversely affect workers' performance. For instance, empirical research in emergency response and call centers illustrates performance deterioration with increased workload (Collewet & Sauermann, 2017). Similarly, gig economy studies highlight that fatigue mediates the relationship between stressful work conditions and declining performance and safety (Useche et al., 2024). However, existing literature typically treats fatigue as a static state and fails to

dynamically capture the interplay between cumulative workload and fatigue-related behavioral adjustments in the context of rapid on-demand deliveries.

1.4 Methodology

This study differentiates itself by explicitly modeling the cumulative nature of workload and its direct implications for fatigue, behavioral performance, and self-regulation by proposing a controlled lab experiment, replicating the key features of quick-commerce operations and delivery environment. Participants are randomly assigned to low, medium, or high delivery targets within timed sessions, with real-time feedback and the option to pause or end their shift. We measure objective performance metrics (delivery times, misses), subjective fatigue ratings, and self-regulation behaviors.

1.5 Contributions

This research advances both theory and practice. Theoretically, we extend behavioral operations models by embedding dynamic fatigue accumulation and self-regulation into gig labor supply frameworks, thereby linking operational efficiency with human factors in on-demand delivery. Practically, our findings will guide quick-commerce firms in designing workload policies—such as mandated micro-rests or dynamic assignment algorithms—to sustain service quality without endangering workers.

2. Literature Review

2.1 Gig Work and Fatigue

Gig economy delivery work is characterized by high uncertainty, irregular hours, and intense time pressure. Unlike regulated trucking or transit jobs, many platforms impose no work-hour limits or formal fatigue management, leaving couriers to self-regulate their workload. The resulting instability in schedules and income is associated with chronic stress. Prolonged stress and erratic hours can lead to fatigue, sleep problems, and declining mental health among gig workers. Over time, these conditions erode well-being and the capacity to maintain fast service. Christie and Ward (2019) documented that *"gig work clearly led some couriers to experience impairment caused by fatigue and pressure to violate speed limits"*, with many admitting to daily near-misses or collisions.

2.2 Workload and Performance Deterioration

Empirical research shows that as work hours or task loads accumulate, human performance tends to decline. In fields like emergency response and call centers, longer shifts consistently degrade productivity. Patterson et al. (2015), performing a study of 743,000 EMS incidents in Mississippi, found that paramedic crews working longer shifts exhibited significantly poorer performance at the end of their shift, an effect attributed to fatigue; the deterioration was large enough to measurably increase 30-day patient mortality by ~0.8%. Similarly, Collewet and Sauermann (2017) studied call center agents and found that increased working hours lead to diminishing returns, especially for less experienced workers, where fatigue significantly reduces productivity. These patterns align with experiments in psychology showing that sustained cognitive or physical effort causes **performance fatigue** – slower reaction times, more errors, and lapses in attention over time.

2.3 Gig Economy Behavioral Studies

Recent behavioral operations research on ride-hailing and delivery platforms provides further evidence that cumulative workload influences labor supply decisions. Allon *et al.* (2023) analyzed hundreds of NYC ride-hailing driver shifts and observed that *"the more workers have earned so far, the more likely they will stop working."* Initially, this was attributed to income-targeting (stopping after hitting earning target), but an alternative explanation is **fatigue**. In fact, the data showed that drivers with longer hours were prone to log off even if they hadn't reached an income target, consistent with fatigue setting in. Farber (2015) similarly found the probability of ending a taxi shift rose sharply with hours worked, more so than with income earned. These behaviors mirror the fatigue–productivity tradeoff: beyond a certain point, each additional delivery or ride comes at a growing personal cost, leading workers to either slow their pace or stop working altogether. In the food delivery context, Useche *et al.* (2024) found that work-related fatigue mediates the relationship between job stressors and safety incidents (traffic crashes). In other words, high workload and time pressure increase rider fatigue, which in turn directly causes more accidents and errors. This underscores fatigue as a "significant yet overlooked threat" – not only for safety, but also for service performance.

3. Model and Hypotheses

3.1 Conceptual Model

To address the central research question —*"How does cumulative workload impact the behavioral performance and self-regulation of gig delivery workers in quick-commerce settings?"*— this study proposes the following behavioral model that maps the dynamic interplay between workload, fatigue, performance, and self-regulation.

Independent Variable: *Cumulative Workload,* operationalized as the number of completed deliveries or total active time during a shift.

Mediator Variable: *Fatigue*, encompassing subjective (e.g. self-reported 1–10 tiredness rating) and objective dimensions (e.g. heart-rate variability).

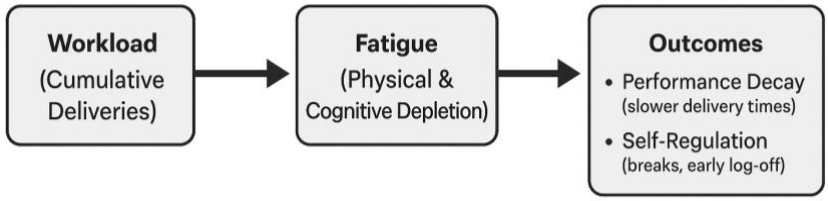
Dependent Variables: Two categories.

(i) Performance Deterioration (higher delivery time, missed deadlines)

(ii) Self-Regulation Behaviors (voluntary breaks, early log-off decisions)

The model postulates a **causal sequence**: as gig workers' cumulative workload increases, so does fatigue, which in turn triggers two observable outcomes: (i) a measurable decline in task performance, and (ii) compensatory self-regulation behaviors intended to mitigate excessive strain. This framework recognizes both direct (workload → fatigue → performance) and indirect (fatigue → self-regulation → reduced output) effects, consistent with behavioral operations and occupational health literature. A visual representation of the model can be seen in **Figure 1**. The mathematical formulations related to the model are in **Appendix**.

Figure 1



3.2 Model Rationale and Assumptions

This model is rooted in the principle that gig workers balance income-maximization against the accumulating disutility of physical and mental exertion (fatigue). Unlike static models which treat fatigue as fixed, our approach captures the *dynamic* buildup of fatigue throughout a shift and recognizes that self-regulation (e.g., taking a break, logging off) is an

endogenous, rational response to escalating fatigue—rather than simply an exogenous dropout or noise. The model has three core assumptions.

→ *Fatigue* accumulates as a non-linear function of workload, possibly accelerating after a threshold.

→ *Fatigue* mediates the relationship between workload and both performance and self-regulation.

→ *Self-regulation actions* (breaks, log-off) are intentional choices in response to perceived fatigue, aimed at optimizing long-term well-being rather than maximizing immediate output.

3.3 Hypotheses Development

H1- Cumulative Workload Increases Fatigue *(Workload → Fatigue)*

"The number of completed deliveries since shift start will show a positive linear association with mean self-reported fatigue (1–10 scale)."

As gig delivery workers complete more orders in a session, their fatigue levels will rise. Each additional delivery (especially under time pressure) incurs physical exertion (driving, carrying goods) and mental effort (navigation, timely responses). We expect a positive relationship between the number of deliveries done (or total time on duty) and the courier's reported fatigue.

H2- Fatigue Deteriorates Delivery Performance *(Fatigue → Performance)*

"For each one-point increase in fatigue at delivery t, the completion time at t + 1 is expected to lengthen relative to the participant's own baseline."

Higher fatigue will negatively impact key performance metrics such as delivery speed and task completion time. When tired, couriers are likely to slow down – for example, riding more cautiously or taking longer breaks between orders – resulting in longer average delivery durations per order. This is a direct consequence of reduced alertness and energy. Prior research found that fatigue led to slower reactions and more errors in both cognitive tasks and driving tasks. In the gig delivery context, Useche *et al.* (2024) showed that fatigue significantly mediated poorer safety outcomes (more crashes) among food couriers. While our focus is on delivery **speed** rather than accidents, the underlying mechanism is similar: a fatigued rider cannot sustain the same pace as a fresh rider. Therefore, we hypothesize that as cumulative deliveries increase (and fatigue builds), the **pace of subsequent deliveries will suffer** – measurable as an increasing delivery completion time trajectory over the shift.

H3- Fatigue Triggers Self-Regulation *(Fatigue → Early Log Off)*

"At any point in the shift, each one-point increase in self-reported fatigue will be associated with a higher hazard of early log-off.

Gig workers who accumulate excessive workload will be more likely to self-regulate by curtailing or pausing their work. The most extreme form of this is **logging off early** (ending the shift earlier then intended). We predict that couriers with a higher prior workload are more prone to voluntarily stop accepting new orders or end their availability for the day. Empirical evidence from ride-hailing supports this notion: drivers frequently choose to stop working after long hours on the road, even if demand (and potential earnings) remain – a pattern attributed to fatigue and diminishing returns. In our context, a rider might plan a 6-hour shift but after, say, 4 hours of non-stop deliveries, decide "that's enough for today" and log off. Even if they do not fully quit, they might start rejecting incoming orders. This hypothesis aligns with self-regulation theory: when individuals feel their performance or safety could be compromised, they consciously down-regulate their effort. We expect to observe a higher probability of early log-off in conditions of high workload.

Taken together, these hypotheses posit a fatigue-mediated pathway from cumulative workload to behavior. We expect our experiment and analysis to demonstrate that *"too much, too fast"* eventually backfires – after a certain point, piling on more deliveries yields slower service and a greater chance that workers drop out early.

4. Experimental Design

4.1 Design Overview & Settings – Treatments – Participant Selection

We implement a between-subjects laboratory experiment that recreates a 30-minute quick-commerce shift. Cohen's power analysis suggests 25–30 subjects per cell yields >80% power to detect medium effects ($d \approx 0.5$) on primary outcomes. Hence, we recruit 90 adults via a crowdsourcing recruitment system. Eligibility conditions are having (i) a driving license, (ii) prior gig-style work or comparable physical job, (iii) self-reported "moderate" fitness level. Participants are randomly allocated (block randomization by session size six) to one of three workload conditions, summarized in Table 1.

Table 1

Condition	Target deliveries in 30 min	Implied pace	Purpose
Low	5	Relaxed	Baseline, minimal fatigue
Medium	10	Steady	Dose–response check
High	15	Continuous	Tests upper-bound fatigue

4.2 Task Mechanics (Step by Step)

The experiment will be performed by following order of tasks.

⇒ **Arrival & Briefing (15 min).** Participants sign consent forms; IDs are coded. We introduce the rules, payoffs and the wearable heart-rate sensor.

⇒ **Dummy trial (3 min).** One dummy delivery ensures interface mastery.

⇒ **Main Task (30 min).** Each simulated "delivery" comprises three sequential actions: (1) participants briskly walk or jog to a marked station in the laboratory, (2) scan a QR code or enter a PIN on a tablet positioned at that station, and (3) return to the dispatch screen to receive their next assignment. A visible countdown accompanies every task; if the allotted time is exceeded, the screen turns red and displays '***Late!'***. To mirror couriers' discretion to self-regulate workload, two on-screen options appear between tasks: *(i) Take a break,* which pauses task assignment for 120 seconds while the session clock continues to run, and *(ii) End shift,* which terminates the session immediately.

⇒ **Post-Task Survey (7 min).** NASA-TLX, manipulation-check items, debrief on pacing strategy will be performed.

⇒ **Payment & Debrief (5 min).** Fixed €5 show-up fee + €0.25 per delivery + €5 completion bonus (if target met) will be paid to participants.

4.3 Performance Metrics

The experimental data set integrates four classes of performance indicators.

- **Objective task performance**—time-stamped completion times for each delivery, deadline adherence (binary), and total number of tasks finished.
- **Behavioral self-regulation**—frequency and duration of voluntary breaks and whether—and when—participants elect to end their shift early.
- **Subjective workload and fatigue**—a brief 1–10 fatigue rating, complemented by post-session NASA-TLX questionnaire
- **Physiological indicators**—continuous heart-rate data captured via wearables, furnishing an objective proxy for physical strain and alertness.

5. Data Analysis

To evaluate our hypotheses, we apply a structured, multi-step analytical workflow; with descriptive statistics to characterize our data, hypothesis testing to assess treatment effects, regression modeling to control for confounders and quantify effect sizes, and exploratory analyses to uncover additional insights.

5.1 Descriptive Statistics

We begin by summarizing key measures across the three workload conditions (low, medium, high). For each condition, we compute the mean and standard deviation of delivery completion times, self-reported fatigue ratings, number and duration of voluntary breaks, and early log-off rates. Trends in fatigue accumulation and performance decay are visualized by line plots (fatigue vs. delivery number) and bar charts (average delivery time and break frequency).

5.2 Exploratory Analysis

Prior to hypothesis testing, we will assess the distribution of each key variable for normality using the Shapiro–Wilk test and by inspecting Q-Q plots. Based on prior literature and scale properties, we anticipate that heart-rate-variability and break counts are likely to be right-skewed and/or zero-inflated, while self-reported fatigue is expected to approximate normality but may show ceiling/floor effects. For variables that meet the assumptions of normality and homogeneity of variance (per Levene's test), parametric tests (paired t-tests, repeated-measures ANOVA) will be used. For variables violating these assumptions, we will employ non-parametric alternatives (Wilcoxon test, Kruskal–Wallis test). This approach ensures our inferential tests are robust to distributional deviations and pre-specifies which variables may require non-parametric methods.

5.3 Statistical Hypothesis Testing

H1: Cumulative workload → fatigue. We perform a one-way ANOVA (or Kruskal–Wallis test if normality is violated) on mean fatigue scores across conditions, expecting a monotonic increase (High > Medium > Low). To examine growth trajectories, we employ a repeated-measures ANOVA (Condition × Delivery Number), testing for a significant Condition × Delivery Number interaction proving steeper fatigue accumulation with high workload.

H2: Fatigue → performance deterioration. Given that multiple deliveries are nested within each rider, we will employ linear mixed-effects models to test the effects of fatigue and workload on delivery performance. These models will include random intercepts for each rider to account for baseline differences, and random slopes for key predictors where justified (e.g., fatigue), capturing heterogeneity in individual responses. This approach enables us to use all available data without aggregation, controls for autocorrelation, and avoids the restrictive sphericity assumption required by repeated-measures ANOVA.

H3: Workload/fatigue → self-regulation. We model the time until log-off using a mixed-effects Cox proportional hazards model with rider-level frailty. Fatigue is entered as a time-varying covariate, allowing us to estimate how each one-point increase on the 1–10 fatigue scale affects the instantaneous hazard of ending a shift.

5.4 Regression Modeling

To control for individual differences (e.g., prior gig experience, baseline fitness) and isolate the effects of workload and fatigue, we estimate mixed-effects regression models, which include random intercepts for participants and fixed effects for condition, cumulative deliveries, and interaction terms. We also examine whether the use of micro-breaks moderates performance decay by including a break-by-time interaction. Physiological indicators (e.g., heart-rate variability) are added as objective covariates to validate self-reported fatigue measures. All mixed-effects regressions include random intercepts and slopes for cumulative workload by participant to capture individual endurance differences, and all figures display 95% CI around estimated trajectories.

Our approach ensures rigorous testing of our hypotheses and affords deeper understanding of how cumulative workload shapes fatigue, performance, and self-regulation in quick-commerce delivery contexts.

6. Conclusion

6.1 Anticipated Findings

Our experimental design predicts three key outcomes. First, participants in the high-workload condition are expected to accumulate fatigue more rapidly than others. Second, this growing fatigue should manifest as a progressive

slowdown in delivery times, confirming a measurable decline in operational efficiency. Third, we anticipate a higher incidence of self-regulation behaviors among riders facing heavier workloads. Collectively, these patterns support the hypothesized fatigue-mediated pathway: excessive delivery pressure eventually diminishes both speed and willingness to continue working.

6.2 Practical Contributions & Managerial Implications

The results can guide quick-commerce platform operators in crafting evidence-based workload policies such as (i) implementing upper limits on sequential deliveries and mandating micro-rests; (ii) deploying real-time assignment algorithms that divert new orders away from riders whose recent load-fatigue indicators are elevated; (iii) integrating wearable fatigue metrics into driver apps to trigger stretch-break prompts; and (iv) redesigning incentives to balance speed with safety—e.g., combining bonuses for on-time delivery windows with rewards for break compliance and sustained quality.

6.3 Theoretical Contributions

This study extends behavioral operations research by embedding dynamic disutility—fatigue—into gig labor-supply models. By illustrating how cumulative workload interacts with self-regulation decisions, we challenge the simplifying assumption of purely income-targeting drivers and demonstrate the value of laboratory replications for disentangling causal mechanisms.

6.4 Limitations

Several constraints temper the generalizability of our findings. The controlled laboratory setting omits real-world hazards such as traffic, weather, and customer interactions, potentially underestimating stress variability. The short, single-session exposure cannot capture chronic fatigue or strategic adaptation over extended periods. Additionally, our sample may differ from full-time couriers in socioeconomic pressures, which could affect endurance thresholds.

6.5 Future research

Future work should include longitudinal field trials that track actual couriers' workload, biometric indicators, and service metrics across multiple weeks; intervention experiments (e.g., enforced break policies vs. control) to quantify the cost–benefit of fatigue mitigation; development of ML models that leverage smart sensors to predict slowdown or in real time; and cross-context validation in ride-hailing, meal delivery, and diverse regulatory environments.

7. References

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8. APPENDIX

Note: We utilized ChatGPT to improve the visualization of mathematical models (variable symbols, font, style etc.)

1. Fatigue-Accumulation Model

We treat time (or delivery index) as a discrete sequence *t = 1, 2, ...* for rider *i*.

$$F_{i,t} = \underbrace{F_{i,t-1}}_{\text{carry-over}} + \alpha_1 \Delta W_{i,t} + \alpha_2 \mathbf{1}(W_{i,t} > \tau)\, \Delta W_{i,t} + u_i + \varepsilon_{i,t}^{(F)}$$

Symbol	Meaning
$F_{i,t}$	Self-reported fatigue (1–10 scale) just **after** delivery *t*
$\Delta W_{i,t}$	Workload increment (usually 1 delivery) in step *t*
$\mathbf{1}(\cdot)$	Indicator capturing a non-linear "fatigue threshold" at cumulative workload τ
u_i	Rider-specific random intercept (baseline endurance)
$\varepsilon_{i,t}^{(F)}$	Idiosyncratic shock

Interpretation.
α_1 measures "normal" fatigue growth per task; α_2 captures the **accelerated** buildup once workload passes threshold τ (consistent with an S-curve). If $\alpha_2 > 0$, fatigue rises faster in later stages of the shift.

2. Performance-Deterioration Model

Delivery completion time (or lateness probability) depends on **current** fatigue and cumulative load.

$$T_{i,t} = \beta_0 + \beta_1 F_{i,t} + \beta_2 W_{i,t} + \beta_3 B_{i,t-1} + v_i + \varepsilon_{i,t}^{(T)}$$

Symbol	Meaning
$T_{i,t}$	Delivery time (sec) or binary lateness indicator
$W_{i,t}$	Cumulative deliveries completed up to *t*
$B_{i,t-1}$	1 if rider took a micro-break just before task *t*
v_i	Random intercept (inherent speed differences)

Interpretation.
β_1 quantifies how each fatigue point slows the next delivery.
β_2 lets workload affect performance **directly** (beyond fatigue).
β_3 tests whether a break immediately restores speed (we expect $\beta_3 < 0$ for time, i.e., faster after a break).

3. Self-Regulation (Log-Off) Hazard Model

We model the *instantaneous* risk that rider *i* ends the shift at elapsed time *t* with a mixed Cox specification:

$$h_i(t) \;=\; h_0(t)\,\exp\!\Big[\gamma_1\,F_i(t) \;+\; \gamma_2\,W_i(t) \;+\; \mathbf{z}_i^{\top}\boldsymbol{\gamma}_3 \;+\; b_i\Big]$$

Symbol	Meaning
$h_i(t)$	Hazard of early log-off at time *t*
$h_0(t)$	Baseline hazard (non-parametric)
$F_i(t)$	Time-varying fatigue (as a continous covariate)
$W_i(t)$	Cumulative workload at time *t*
$\mathbf{z}_i$	Static controls (experience, fitness, gender)
b_i	Gamma-distributed frailty (unobserved heterogeneity)

Interpretation.
The exponentiated coefficients give fatigue-elasticities of quitting. For example, $\exp(\gamma_1) = 1.25$ means each extra fatigue point raises the **instantaneous** quitting risk by 25 %.